



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE215: Data Structures

Batch: 28th

Date: 08/06/2026

Mid Term Examination

Semester: Spring 2026

Time: 1 hour 30 minutes

Total Marks: 30

Answer any **five** questions from the following:

5x6= 30

- | | | | |
|---|--|-------|---|
| 1 | (a) Define data structure. Classify data structures into primitive and non-primitive types with examples. | CLO1 | 3 |
| | (b) Explain how arrays, linked lists, stacks, and queues can be applied effectively in solving programming problems. | CLO 1 | 3 |
| 2 | (a) Given Array A = {35, 12, 9, 48, 21, 5, 17, 30, 26}, apply the Merge Sort algorithm to sort the elements in ascending order. | CLO2 | 3 |
| | (b) Summarize the time complexities of Merge Sort, Selection Sort, and Bubble Sort in best, average, and worst cases. | CLO2 | 3 |
| 3 | (a) Perform Bucket Sort on the following list: {23, 45, 12, 67, 34, 89, 5, 78, 56, 90}. Assume each bucket covers a range of 10 values. | CLO2 | 3 |
| | (b) Sort the array {64, 34, 25, 12, 22, 11, 90} using Bubble Sort. Show the intermediate steps after each pass. | CLO2 | 3 |
| 4 | (a) Compare Binary Search and Ternary Search in terms of number of divisions, time complexity, and practical efficiency. | CLO 4 | 3 |
| | (b) Apply Binary Search Algorithm to find the element 28 in the sorted array [4, 8, 12, 16, 20, 24, 28, 32, 36]. Show each step clearly. | CLO2 | 3 |
| 5 | (a) Apply Selection Sort algorithm to the array A = {64, 25, 12, 22, 11, 90, 45}, arranging the numbers in ascending order. | CLO2 | 3 |
| | (b) Elaborate on the concept of the Divide-and-Conquer strategy. | CLO1 | 3 |
| 6 | (a) Convert the following infix expression to a Binary Tree: $A*(B+C)-D/E+F$. | CLO4 | 3 |
| | (b) Explain the distinctions between overflow and underflow. | CLO1 | 3 |
| 7 | (a) List out the applications of Linked Lists. | CLO1 | 2 |
| | (b) Given a singly linked list with the elements {6, 12, 18, 24}, insert a new node with the value 15 at the 3rd position in the list. | CLO1 | 2 |
| | (c) Given a singly linked list with the elements {5, 10, 15, 20, 25}, delete the node at the 4th position and return the updated list. | CLO1 | 2 |

BODMAS



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology
Program: B.A. (Hons)
3rd Semester Examination

Course Code: BAN211

Course Title: History of Emergence of Bangladesh

MidTerm Examination

Date: 13.06.2026

Total Time: 1.5 Hours

Semester: Spring 2026

Total Marks: 30

Instruction & Information

- Answer any *three* questions
- All questions carry equal marks

1. Describe the socio-economic and cultural development during the Sultanate Period. 10
2. Describe the socio and cultural development during the Mughal Period. 10
3. Mention the naming of Bengal and Border of Bengal during the Muslim Sultanate. 10
4. Write the Biography of Shamsuddin Md Iliyas Shah and Ikhtiyar-ud-din Md Bakhtiyar Khilji. 10



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science (Honours) in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT217

Course Title: Ordinary and Partial Differential Equations

Midterm Examination

Semester: Spring 2026

Date: 6/20/2026

Total Time: 1:30 Hours

Total Marks: 30

3 × 10 = 30

Answer any three (03) out of the following questions:

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No	Question	CLOS	Marks
Q1.	(a) Define the order and degree of an ordinary differential equation. State the conditions under which an ODE is classified as linear. Determine the order, degree, and linearity of $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = k \frac{d^2y}{dx^2}$	CLO1	[4]
	(b) Clearly distinguish between a particular Solution and a general Solution. Can a singular solution be obtained by assigning specific values to the arbitrary constants of a general solution? Justify your answer.	CLO2	[2]
	(c) Form the ordinary differential equation by eliminating the arbitrary coefficients constants A and B from the following relation $y = Ae^{3x} + Be^{-2x}$.	CLO3	[4]
Q2.	(a) Solve the non-linear differential equation by reducing it to a variable-separable form using an appropriate substitution $\frac{dy}{dx} = \tan^2(x + y)$.	CLO2	[4]
	(b) Define a homogeneous differential equation of the first order. Show that the following equation is homogeneous and find its general solution $\left(x \cos \frac{y}{x} + y \sin \frac{y}{x}\right) x dy = \left(y \sin \frac{y}{x} - x \cos \frac{y}{x}\right) y dx$	CLO3	[6]
Q3.	(a) State the necessary and sufficient condition for the differential equation $M(x, y)dx + N(x, y)dy = 0$ to be exact. Hence, solve the following exact equation: $(2xye^x + y^2e^x)dx + (2xe^x + y)dy = 0 ; y(0) = -2$	CLO2	[5]
	(b) Show that the differential equation $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$ is not exact. Find an appropriate integrating factor $u(y)$ to make it exact, and find its general solution.	CLO3	[5]
Q4.	(a) What do you mean by initial and boundary value problem?	CLO1	[2]
	(b) Find the general solution of the following linear differential equation: $(1 + x^2) \frac{dy}{dx} + 2xy = \cos 2x$.	CLO3	[4]
	(c) Solve the initial value problem (IVP) $x \frac{dy}{dx} - y = x^2e^x ; y(1) = e$.	CLO3	[4]

- Q5. (a) What is a Bernoulli Equation? CLO1 [1]
- (b) Reduce the following equation into linear form and find its complete primitive (general solution) $x \frac{dy}{dx} + y = y^2 \ln x$. CLO2 [3]
- (c) Find the general solution of the following first-order non-linear differential equation $\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos^2 y$ for $x > 0$. CLO3 [3]
- (d) Solve the following second-order differential equation $\frac{d^2y}{dx^2} = 6x$, satisfying the boundary conditions $y(0) = 2$, $y'(1) = 5$. CLO3 [3]



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science (Honours) in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT213

Course Title: Statistics & Probability

Midterm Examination

Semester: Spring 2026

Date: 6/15/2026

Total Time: 1:30 Hours

Total Marks: 30

Answer any three (03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No	Question	CLOS	Marks
Q1.	(a) Define index number.	CLO1	[2]
	(b) Write down the characteristic and uses of index number.	CLO2	[3]
	(c) Calculate the price index from the following table by using Fisher's Ideal Method and also check revel test. .	CLO3	[5]

Commodity and unit	Price (in tk) 2006	Price (in tk) 2007	Quantity 2006	Quantity 2007
Rice (kg)	55	40	12	14
Oil (liter)	75	90	16	20
Dal (kg)	60	78	6	4
Milk (liter)	35	44	10	15
Meat (kg)	200	240	8	6

Q2.	(a) Show that the value of correlation co-efficient is -1 when $y = a - bx$.	CLO1	[2]
	(b)	CLO3	[8]

Table 1	
M	N
20.25	25
27	30.75
29	31.25
32.5	28
37.75	40
39	15

Table 2	
P	Q
10	25
12	17.75
15	23
20	21
23	14
27	10.5

- Calculate the correlation co-efficient of table 1.
- Draw scatter diagram.
- Identify and write down its relationship.
- Compare Table 1 and Table 2.

- Q3. (a) What do you mean by random variable and discrete random variable. CLO1 [2]
 (b) Consider the sample space with children of a family. Suppose the probability of boy is $\frac{2}{5}$. Let X be the number of boys. Three children are selected. Then calculate- CLO3 [5]
 (i) Exactly two boys
 (ii) At most two boys
 (iii) No boys
 (c) Compute the probabilities of the size of the shirt sold from the given table and also calculate more than M, and between L-XXL. CLO3 [3]

Size of shirts	S	M	L	XL	XXL
No. of pair sold	9	13	12	8	18

- Q4. (a) Prove that the correlation co-efficient lies between -1 to +1. CLO1 [3]
 (b) Compare the price index from the following table: CLO3 [7]

Commodities	2023		2025	
	Price	Quantity	Price	Quantity
I	20	16.25	30	11.75
II	17.75	7	23	7.25
III	12	5	7	14
IV	9	1.5	25.5	4

- Q5. (a) What do you mean by probability density function? CLO1 [2]
 (b) (i) If $f(x) = K \sec^2 x - \cos 2x$ $0 \leq x \leq 150$ then CLO3 [8]
 (ii) Find the value of K.
 (iii) Find the probabilities at 0 to 45, at 60, and between 135 and 150.



HAMDARD UNIVERSITY BANGLADESH

Department of CSE

Program: BSc in CSE
Mid-Term Examination- Spring 2026

Course: ACT-215: Financial and Managerial Accounting

Time: 1 Hour 30 Minutes

Total Marks: 30

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answer to questions should be neat and clean.
- Answer any **Three (03)** of the following questions.

No.	Question/ Statement	CLOs	Marks
1.	(a) Define accounting? (b) Briefly describe the users of accounting information. (c) Explain the Historical cost principles and the business entity assumption.	CLO 1	2 4 4
2.	(a). Describe the basic steps in the recording process. (b). Hillen Meyer was started on April 1, 2026 by F. L. Wright and associates. The following selected events and transactions occurred during April. Apr. 1. Hillen Meyer invested \$35,000 cash in the business. 4. Purchased land costing \$27,000 for cash. 8. Incurred advertising expense of \$1,800 on account. 11. Paid salaries to employees \$1,500. 12. Hired Park manager at a salary of \$4,000 per month, effective May 1. 13. Paid \$1,650 cash for a one-year insurance policy. 17. Withdrew \$1,000 cash for personal use. 20. Received \$6,800 in cash for admission fees. 30. Paid \$900 on balance owed for advertising incurred on April 8. Instructions; Journalize the April transactions	CLO 1,3	2 8
3.	(a) Define transaction. State the features of transactions. (a) Indicate whether each of the statements presented below is transactions or events. 1) Depreciation of a company vehicle over time tk.5,000 2) A customer placing an order for goods which value tk.1,500. 3) An owner withdrawing cash for personal use tk.500. 4) Owner met with supplier for purchasing materials which worth tk.1,50,000 5) Sold old printer for tk.3,000 6) Signing a major contract with a new client 7) Bought stock worth tk.25,000 8) Paid rent of tk.15,000 9) Paying an employee's salary tk.30,000 10) A competitor launching a new product	CLO 1,2	5 5

4	Kinney's Repair Ltd. was started a business on May 1, 2025. A summary of May transactions is presented below. 1. Owner's invested tk.10,000 cash in the business. 2. Purchased equipment for tk.5,000 cash. 3. Paid tk.400 cash for May office rent. 4. Paid tk.700 cash for supplies. 5. Incurred tk.250 of advertising costs in the Beacon News on account. 6. Received tk.4,700 in cash from customers for repair service. 7. Owner's withdraw tk.1,000 cash for personal use 8. Paid part-time employee salaries tk.1,000. 9. Paid utility bills tk.160. 10. Performed repair service worth tk.980 on account. 11. Collected cash of tk.120 for services billed in transaction (10). Instructions Prepare a tabular analysis of the transactions, using the following column headings: Cash, Accounts Receivable, Supplies, Equipment, Accounts Payable, Capital, Revenues, Expenses, and Drawing.	CLO 2	10
5	Nipun opened a business name 'Nipun enterprise' on December 1, 2025. The following selected events and transactions occurred during the month of December 2025. Dec-01: Invested cash by the owner Tk.50,000 and an old motor bike valued Tk.30,000. Dec-03: Computer purchased Tk.20,000; cash payment Tk.8,000 and rest on credit. Dec-04: Purchased goods for Tk.60,000 in cash. Dec-05 Purchased supplies Tk.60,000 in cash. Dec-06: Sold goods Tk.80,000 in cash. Dec-10: Cash payment Tk.7,000 for December 03, transaction. Dec-15: Paid for rent expense Tk.1,500. Dec-20: Additional investment by owner Tk.20,000. Dec-30: Drawing made by the owner Tk.500. Dec-31: Stores wages paid Tk.500. Required: a. Journalize the transactions. b. Post the transactions into the ledger uses the following accounts: Cash, Capital, Supplies, Wages Expense.	CLO 3	10

6560, 860, 700, 5000 | (250), 10000, 5680, (1810), (1000)
 13120
 14250
 12680
 112870



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE213: Digital Electronics

Batch:28th

Date:17/06/2026

Midterm Examination

Semester: Spring 2026

Time: 1 Hour 30 Minutes

Total Marks: 30

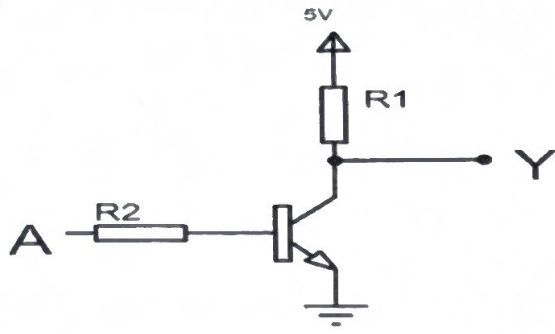
Answer any **five** questions.

5 × 6 = 30

1. (a) Define logic gates. Describe the basic logic gates and provide their truth tables and equations. CLO 1 3
(b) A logic gate produces a HIGH output when its two inputs are different. Identify the logic gate and describe its operation along with its truth table. CLO 1 3
2. (a) How can a **NOR** gate be implemented using only **NAND** gates. CLO 1 3
(b) A boolean function is given by:
 $Y=AB+C$
i) Identify the basic gates required to implement the given function.
ii) Implement the given function using only **NOR** gate. CLO 1 3
3. (a) A boolean function is defined as:
 $Y=A.B$
i) Implement the given boolean function using diode logic.
ii) Can the same logic be extended to three variables using diode logic? Justify your answer. CLO 3 1+2=3
(b) Is it possible to implement a **NOT** gate using diode logic? Explain your reasoning. CLO 3 3
4. (a) Define and describe the following terms with proper examples.
i) SOP form ii) POS form iii) Canonical form CLO 1 3
(b) Simplify the following boolean function using a Karnaugh map (K-map):
 $f(A,B,C,D)=\sum(3,5,7,8,10,11,12,13)$ CLO 1 3
5. Simplify the following equation and implement the simplified equation using **NOR** gate. CLO 1 6
 $f(A,B,C,D)=\sum m(0,1,2,7,8,10)+d(3,5,6,11,15)$
6. Write the following function in canonical SOP form, simplify the function and draw the truth table: CLO 1 6
 $f(A,B,C,D)=\sum(0,1,2,4,5,6,8,9,10,11,12)$

7. Observe the diagram and answer the following questions.

CLO 3 2+2+2
=6



- i) Identify the logic family used in the given diagram.
- ii) Describe the given logic gate with truth table.
- iii) Using the identified logic family, implement a 4-input **OR** gate.